AE 452 Soil and Water Engineering II (2,3,3)

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Agro-hydrology

Agro-hydrology

Learning Objective

1. To revise with students the hydrological cycle and its components
2. To develop analytical skills relevant in the areas of precipitation, evapotranspiration, infiltration and streamflow

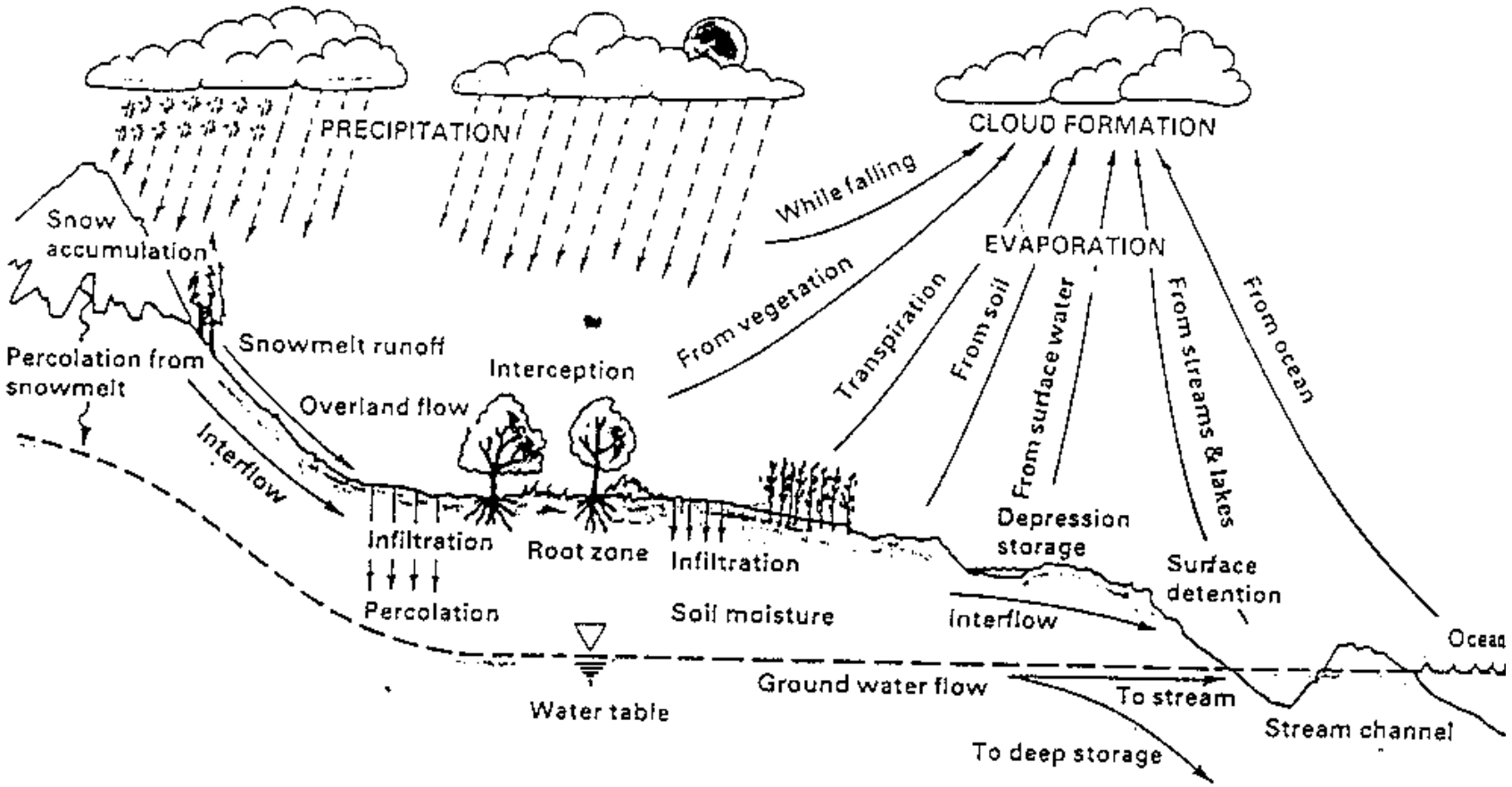
Agro-hydrology

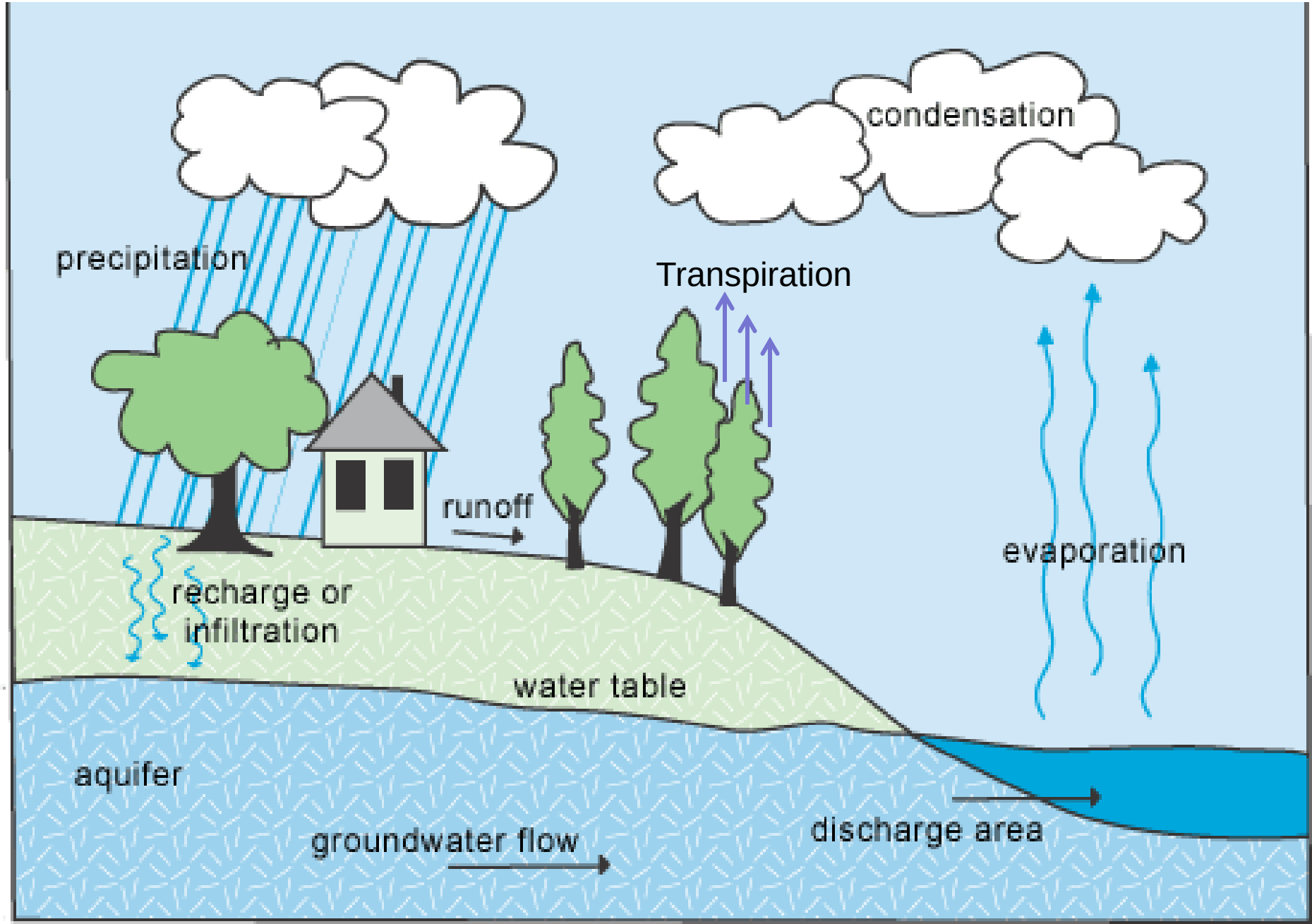
Definition:

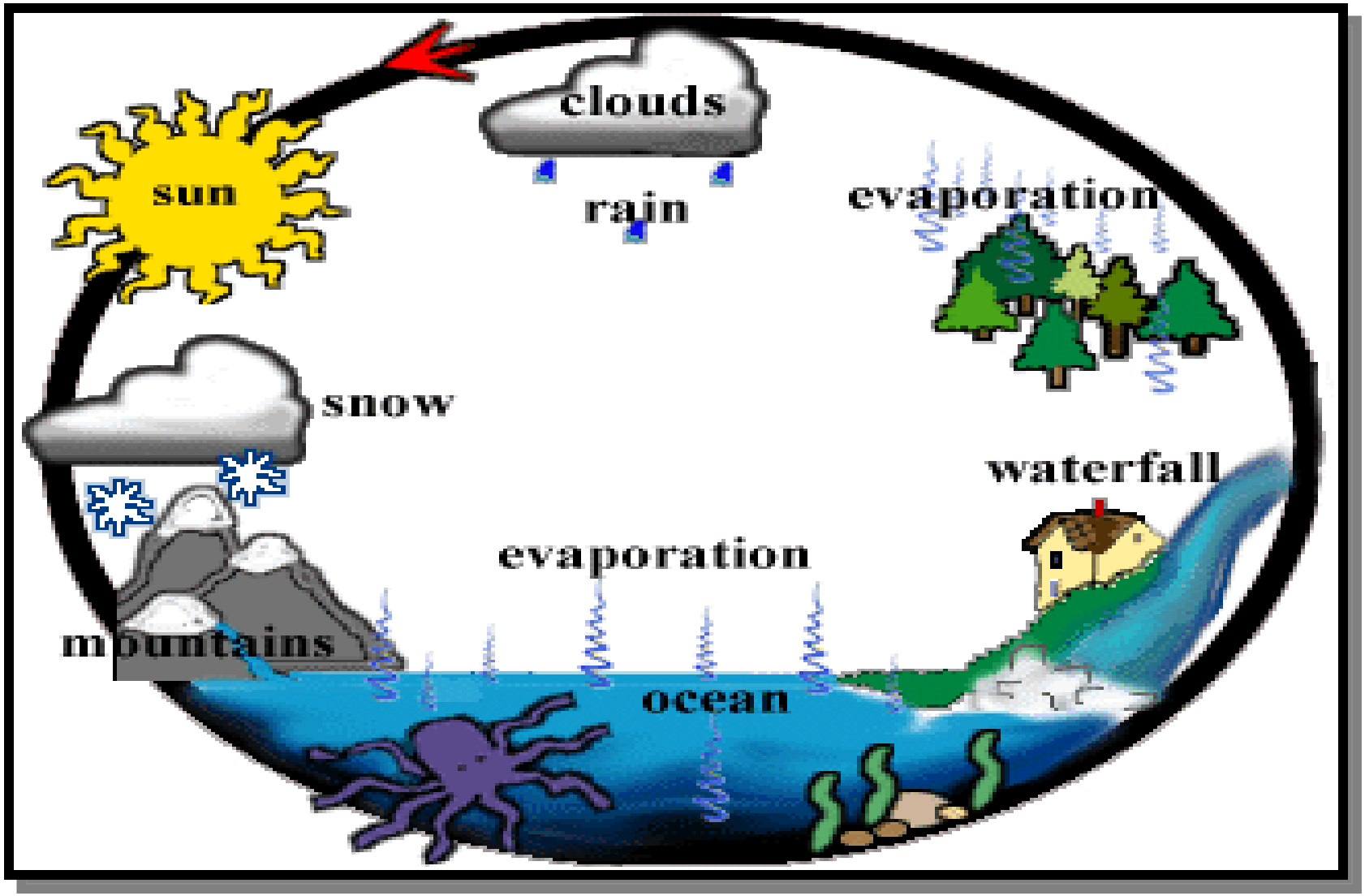
Is the study of water movement, distribution, quality, use and conservation on earth and in the atmosphere as it relates to agriculture.

The importance of **agro-hydrology** can be illustrated with the use of the hydrological cycle.

Hydrologic cycle




condensation
precipitation
Transpiration
runoff
evaporation
recharge or
infiltration
water table
aquifer
groundwater flow
discharge area


clouds
sun
rain
evaporation
snow
waterfall
evaporation
mountains
ocean

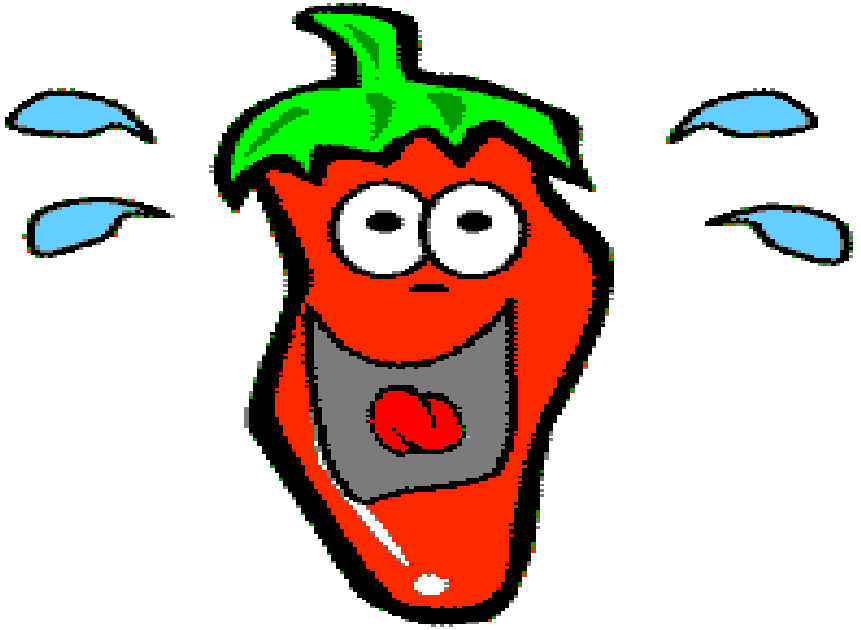
EVAPORATION

Evaporation is when the sun heats up water in rivers, lakes, ocean and or soil and turns it into vapour or steam which rises in to the air.



TRANSPIRATION

Do plants sweat? Kind of. Transpiration is the process by which plants lose water from their leaves. The water rises in to the air.



CONDENSATION

Water vapour in the air gets cold and changes back into liquid, forming clouds. This is called condensation.



PRECIPITATION

Precipitation occurs when so much water has condensed that the air cannot hold it anymore. Water falls to the earth in the form of rain, hail, sleet or snow.



Precipitation

Precipitation types:

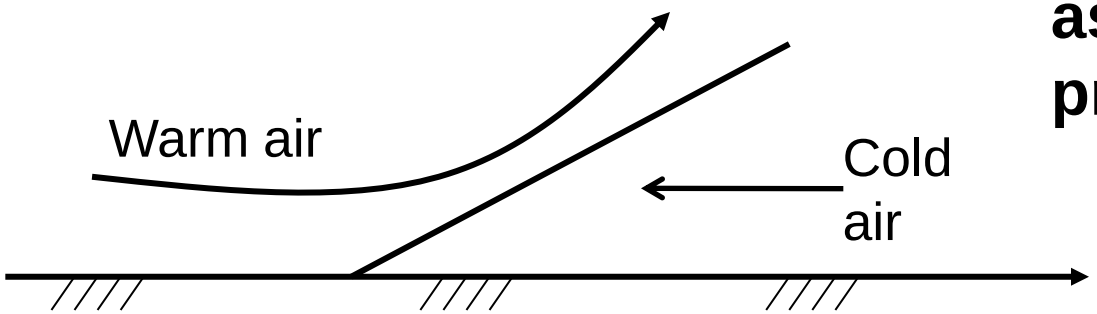
- Falling water droplet such as:
  - **rain**
  - **drizzle** (uniform precipitation with droplet sizes less that 0.5 mm)
- Frozen water particles such as:
  - **snow**
  - **sleet** (raindrops fall through below freezing air temperature resulting in a mixture of rain and snow or hail)
  - **hail** (pellets of frozen ice larger than 5 mm)
- Condense water:
  - **dew** (formed from the direct condensation and absorption from the atmosphere)

Primary steps for formation of precipitation

1. Creation of saturation conditions in the atmosphere (i.e. typically when there is lifting of air mass)
2. Phase change from vapour to liquid (cloud formation)
3. Growth of droplets to appreciable size that will be able to overcome upward air movement

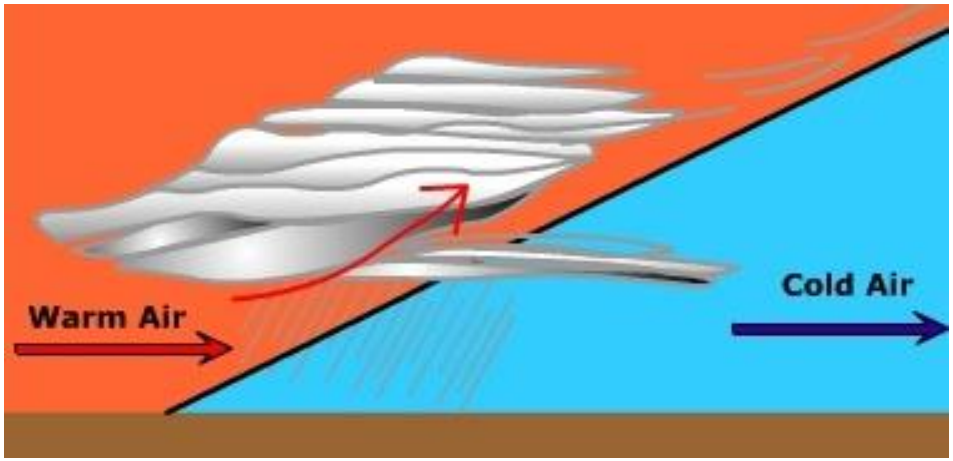
CREATION OF SATURATION CONDITIONS
Lifting Mechanisms (1/4)

1. **Warm Front**: Warm air advancing on cold air



Characteristics of associated precipitation events:

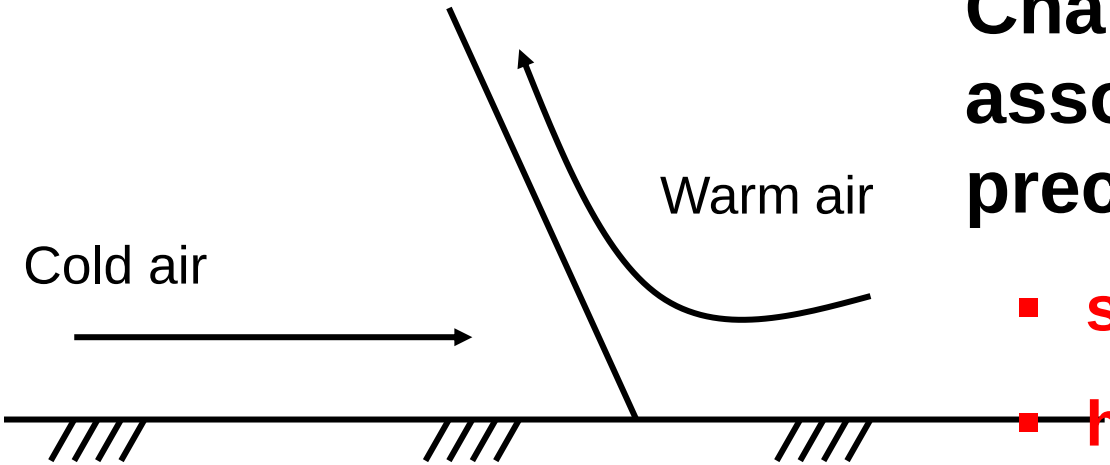
- **longer duration**
- **lower intensity**
- **low energy**
- **gentler storms**



Warm front (produces stratus cloud)

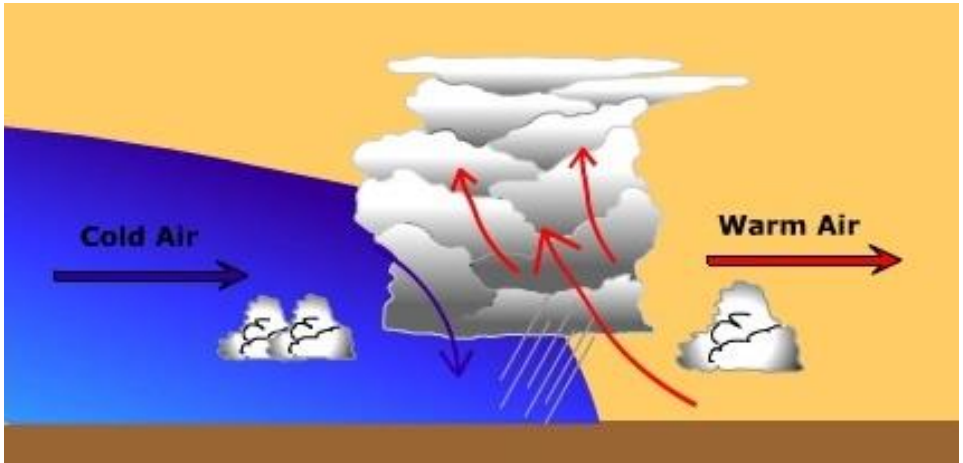
Lifting Mechanisms (2/4)

2. **Cold front**: Cold air advancing on warm air



Characteristics of associated precipitation events:

- **shorter duration**
- **higher intensity**
- **high energy**
- **severe storms**



Cold front (produces cumulus cloud)

Lifting Mechanisms (3/4)

3. Cyclonic Events

- *Tropical Cyclones*: These are non-frontal events but convective and generated by rising of warm-moist air masses.

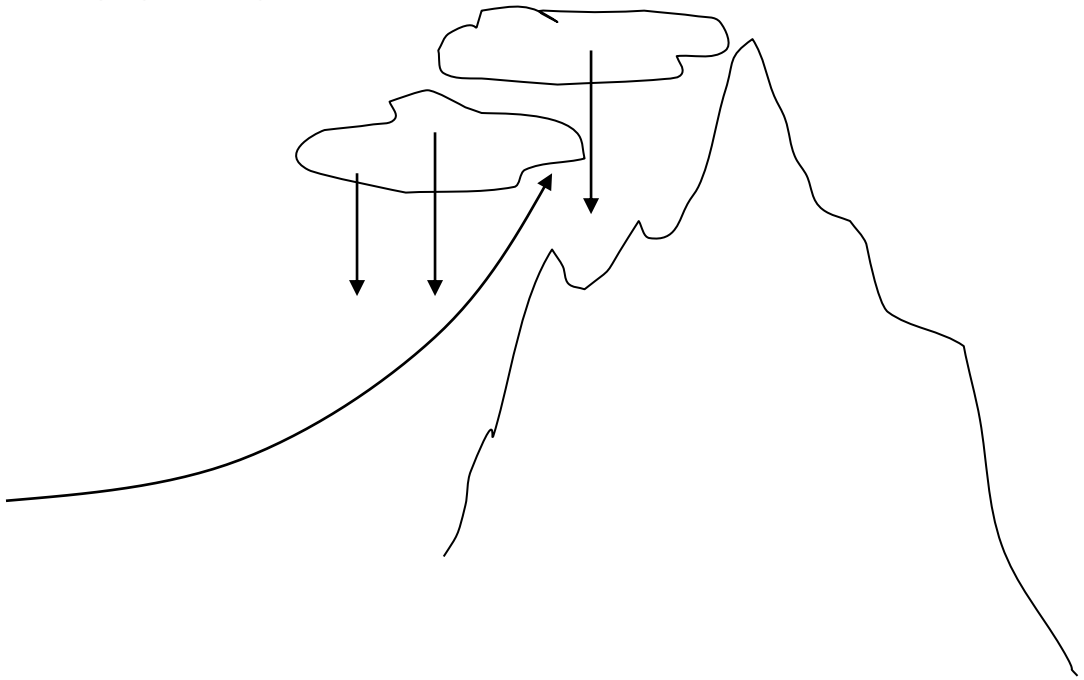
  Typical requirements:

  - Intense sunshine (typical of ocean)
  - High humidity and temperature (typical of ocean)
  - Low surface friction (typical of ocean)

- *Monsoons*:
  - 180° reversal in prevailing wind direction with season.
  - Partly due to differential heating of oceans and continents (India, West Africa), thus bringing a season of wet weather
  - Most prominent when combined with topographic effect that help to create a semi-permanent low pressure over land, drawing moisture from the ocean

Lifting Mechanisms (4/4)

4. **Orographic Cooling (lifting):** Cooling of air as it rises over mountain range. In this case a greater percentage of the precipitation occurs on the windward side of the mountain



Change of Phase (nucleation)

Once saturated conditions exist droplets can be formed, however surface tension resists nucleation.

The presence of small particles (aerosols) provide the initial condensation nucleus for droplet formation.

Condensation Nuclei

1. Most common is **sea salt aerosols**
2. Pollutants such as **dust** and **combustion by-products**
3. **Ice** also enhances formation due to the difference in vapour pressure of water versus ice.

Droplet Growth

- Principal growth mechanism is coalescence or growth by collision.
- Collisions are generated by turbulent air motion and falling rain drops.
- A droplet must grow to a size large enough to overcome the upward convective velocities.

Cloud Seeding

Def.: Artificial nucleation to induce precipitation.

- Rainfall may be induced by the introduction of artificial nucleation materials (such as **Silver iodide** or **dry ice**) into the atmosphere.
- The long term effect of such activity on rainfall variability is not clearly known, as extreme events or events that would have occurred without seeding is difficult to evaluate

Rainfall measurement

Rain gauges are instruments used to measure the **depth** and or **intensity** of falling rain on flat surfaces.

- Factors that affect rainfall measurement:
  - design of the rain gauge
  - topography of location
  - presence of obstruction (such as trees and buildings).
- Main types of rain gauges:
  - non-recording
  - recording types

Non-recording Gauges

- It is used to record the **rainfall amount over a period of one day**. Typically it is an open receptacle leading to a graduated or non-graduated cylinder.
- With a non-graduated cylinder a dipstick or measuring cylinder is used to take the measurement.
- Characteristics
  - Standard rain gauge is for daily (24 hour) rainfall
  - It is economical
  - requires service after the rains
  - relatively free of maintenance

Non-recording Gauges

- Main sources of error may arise from:
  - measuring stick
  - leaks in the funnel or can
  - denting of can
  - evaporation

Recording Gauges

- **Functions**: Provides automatic recording of rainfall depth over time interval, therefore rainfall intensity
- In older gauges drum strip chart was used as the recording mechanism. Modern gauges use digital recording mechanisms.
- **Types of mechanism**
  - weighing bucket
  - tipping bucket
  - float-siphon type
- Adv.
  - suitable for poorly accessible areas
  - records the rainfall amount with time.

Sources of error in rainfall measurement

Sources of measurement error that can lead to error in estimating the actual precipitation:

- observational error
- instrument error
- raindrop splash
- equipment condition
- wind effect is greatest (most significant) on snow and light rain.

Radar rainfall measurement:

- Signals of reflected radiation from raindrop particles at high frequency gives an indication of rainfall. The signal is converted to cumulative rainfall.
- The data can provide information on location and movement of storms and data in ungauged areas.
- However, there are difficulties or error in the measurements due to various types of particulate (hail, snow, rain)

Rainfall Variability

- **Spatial variability – variation from one location to another**: It is generally heaviest near the equator decreasing with increasing latitude with regional geographic location variation having a strong influence
- **Temporal variability – variation from one period or time to another**
  1. Mean variability: Variability in annual or seasonal mean (rainy season to the next rainy season)
  2. Seasonal variability: This is the variation from one season to the other eg. Rainy to dry season. Seasonal variability is very important in the construction of drainage and storage units.
  3. Individual storm variability or the time distribution of individual storms.

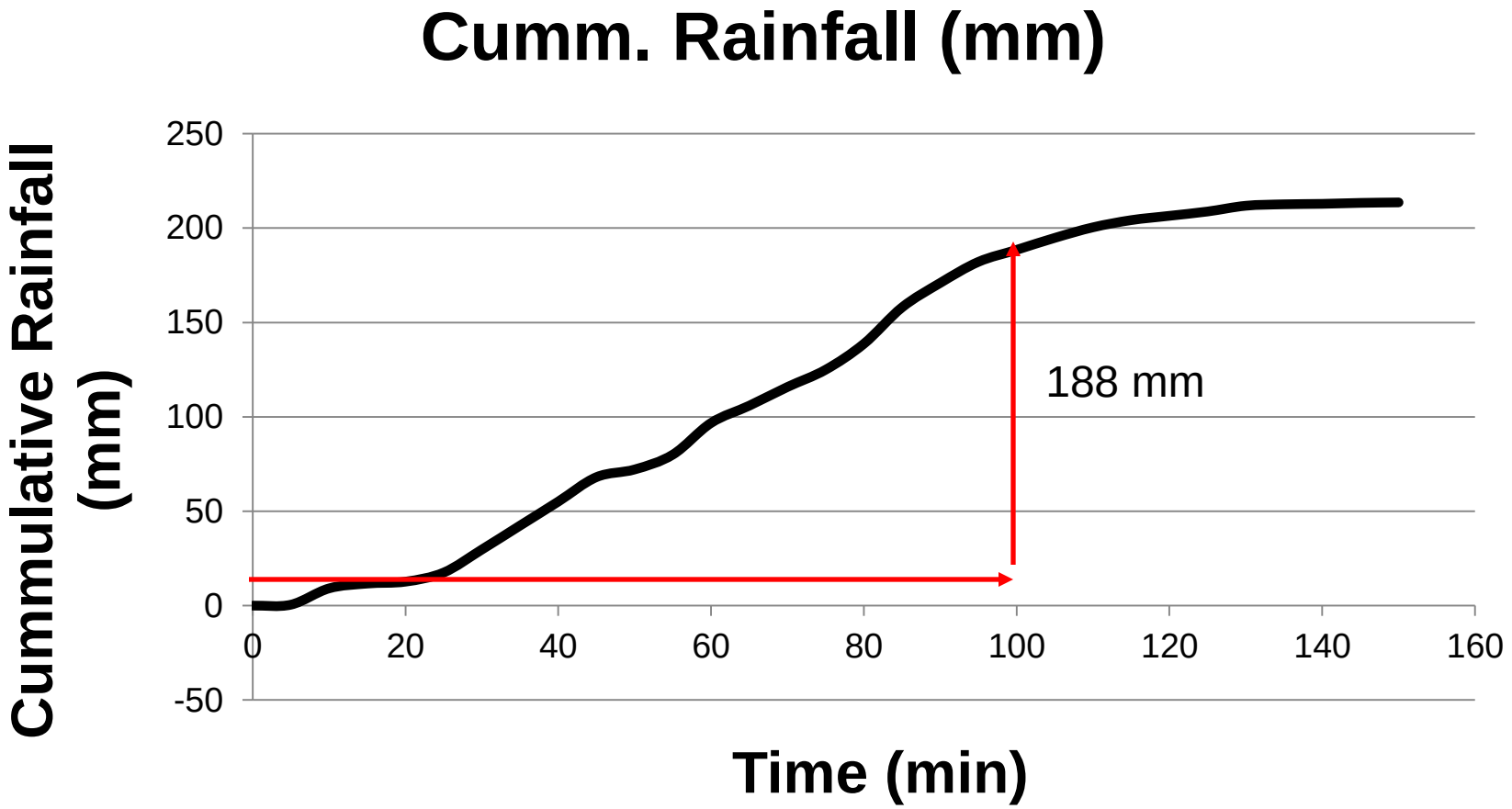
Rainfall data

- The important parameters of rainfall are intensity, duration and frequency.
- **Intensity:** is the amount of rainfall per unit time (usually expressed in mm per hour)
- **Duration**: is the time period for a given storm
- **Frequency**: The number of times a given storm may occur within a period
- The period taken for a storm of a certain intensity and duration to re-occur is referred to as the return period or recurrence interval.

Rainfall data representation

1. Cumulative mass curve

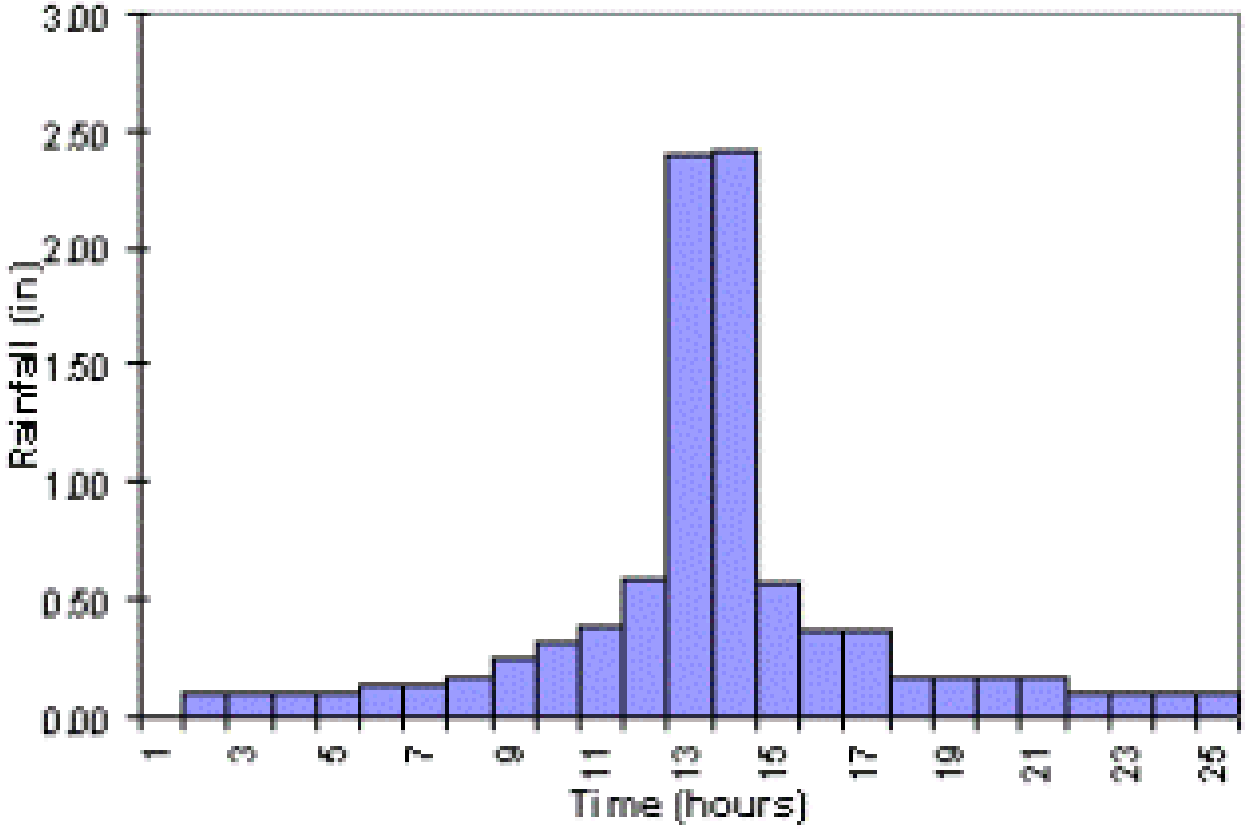
Plot of summation of rainfall increments as a function of time
Typically the output from a recording gauge is in the form of cumulative rainfall over time.



Rainfall data representation

2. Rainfall hyetograph

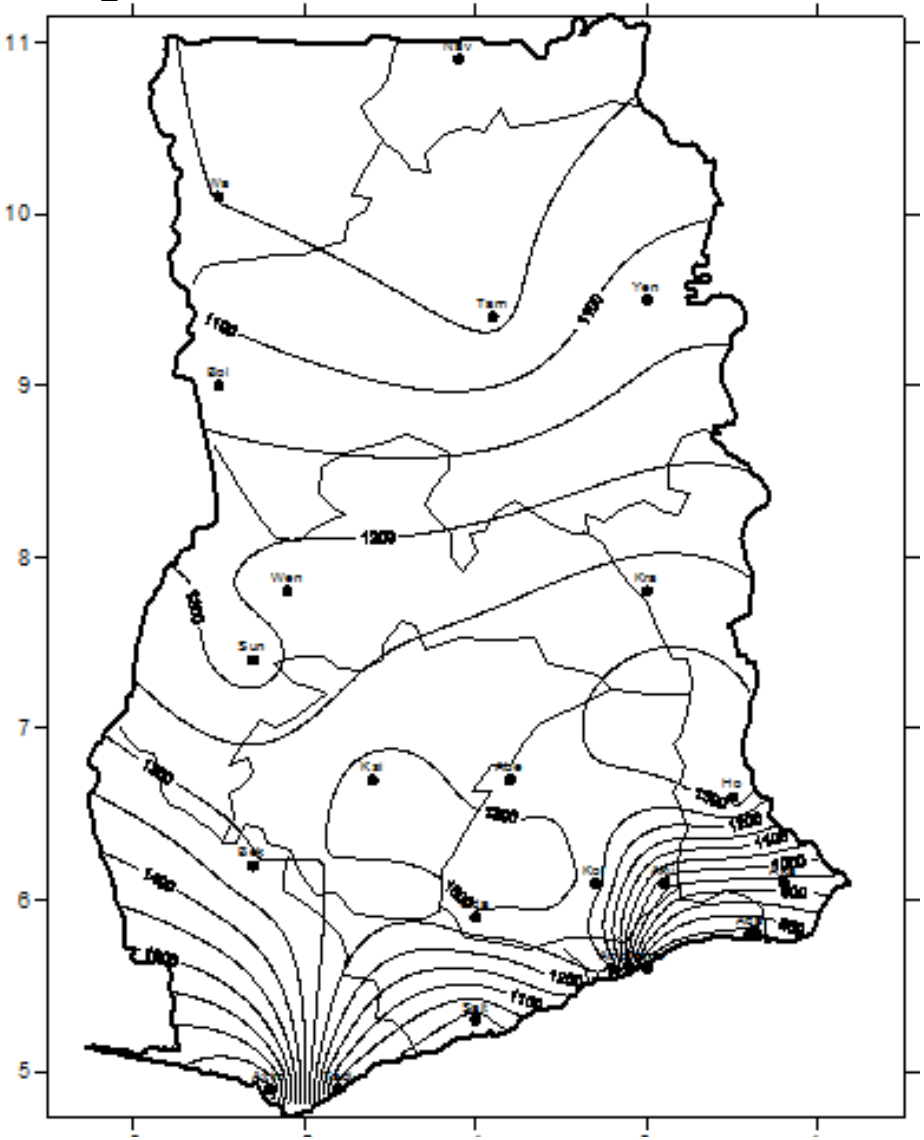
The depth or intensity of rainfall plotted against time interval to generate a hyetograph characterizing the time distribution of a storm or selected time period. Important input to hydrologic models (surface hydrology models).



Rainfall data representation

3. **Isohyetal maps:**

   is a plot of constant rainfall depth (sometimes intensity) contours based on rainfall data recorded at gauged locations (points). These are generated similarly to topographic contour maps. Isohyetal maps are used to represent long-term averages or individual storm events.



Isoyetal map of Ghana

Point rainfall data assessment

Most often rainfall data is measured at a point, which is used for the following:

- Estimate average rainfall over large area (watershed)
- Reduce point rainfall to aerial rainfall for model applications to large areas
- Estimate missing record of gauge data or estimate data at an ungauged location

Estimation of Precipitation over an Area

1. **Arithmetic Averages**: *Used when*
   - *gauges are uniformly distributed*
   - *individual gauge measurements do not vary greatly (typically for small areas)*

$$Pavg = \frac{\sum_{i=1}^{n} P_i}{n}$$

P_i = rainfall at gauge i
n = number of gauges

2. Thiessen Polygon

- Underlying concept is rainfall for an area is represented by the nearest gauge
- This is done by constructing polygons around each gauge.
  - First, by drawing lines to connect each gauge.
  - Then draw perpendicular bisector on each connecting line.
  - Connect bisectors to form polygons
  - Determine area for each polygon

Thiessen Polygon

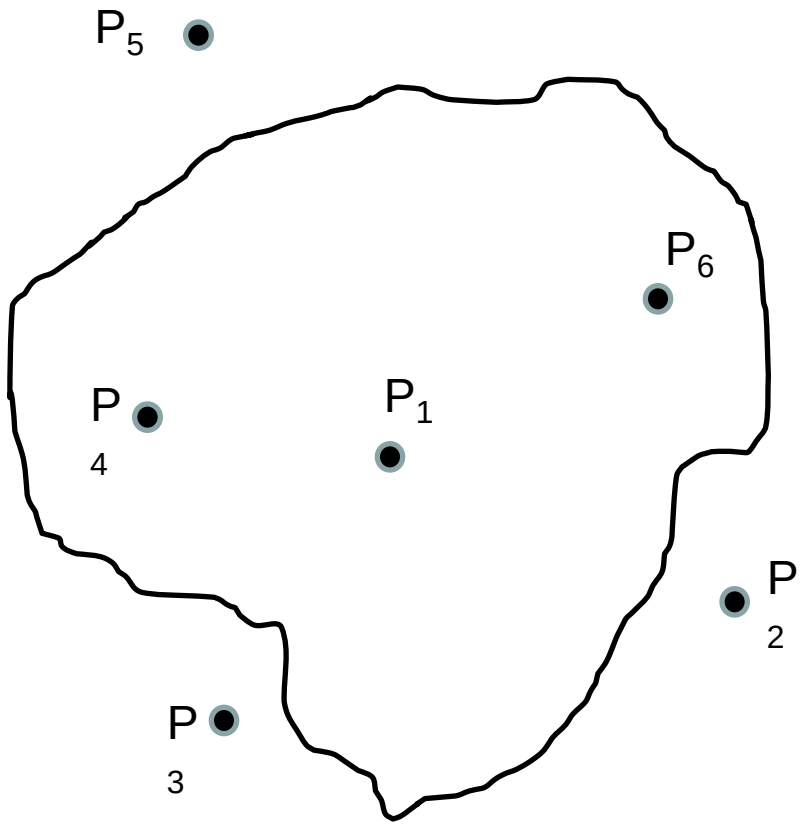
$$Pavg = \frac{\sum_{i=1}^{n} A_i P_i}{A_T}$$

n = number of polygons (gauges)
A_T = total area
A_i = area of polygon around gauge i
P_i = rainfall at gauge i

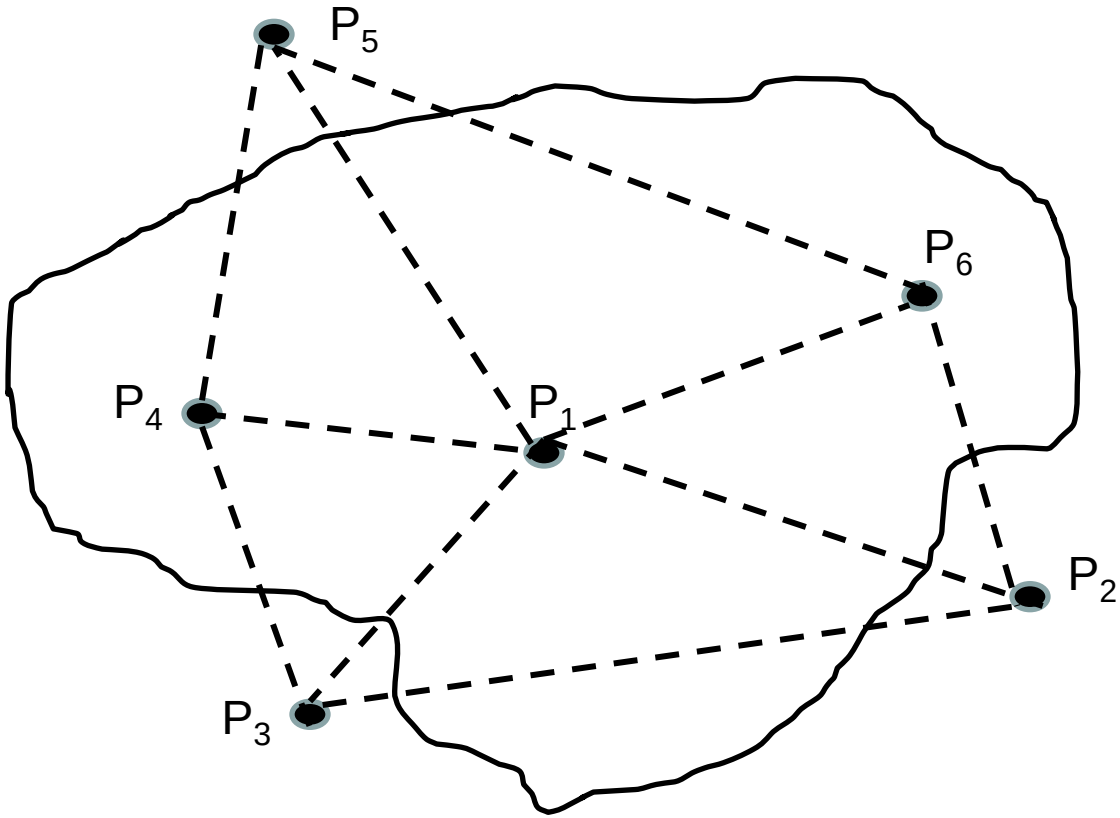
Sample calculation: Arithmetic Average and Thiessen polygon

Determine the mean rainfall using the arithmetic average and the Thiessen polygon methods

Rain gauge	**Precipitation (P_i)(mm)**
P_1	56
P_2	40
P_3	44
P_4	52
P_5	48
P_6	52
Total	

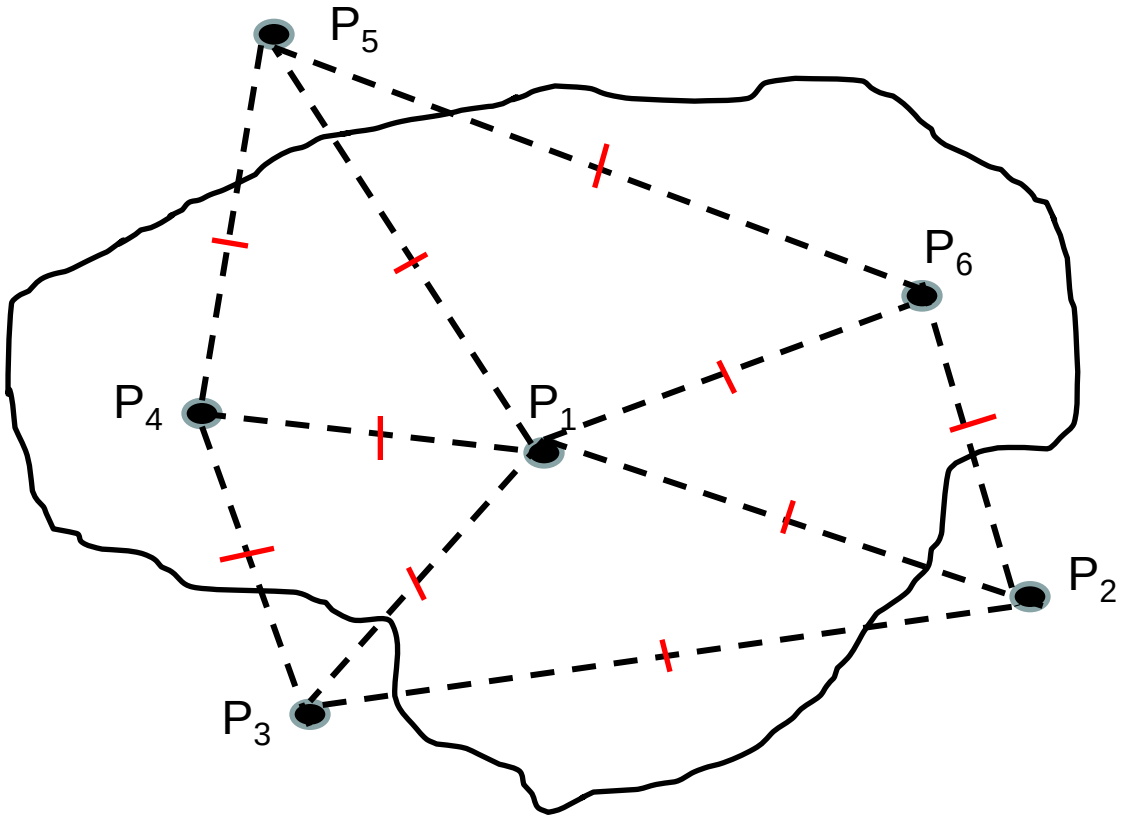


Thiessen polygon



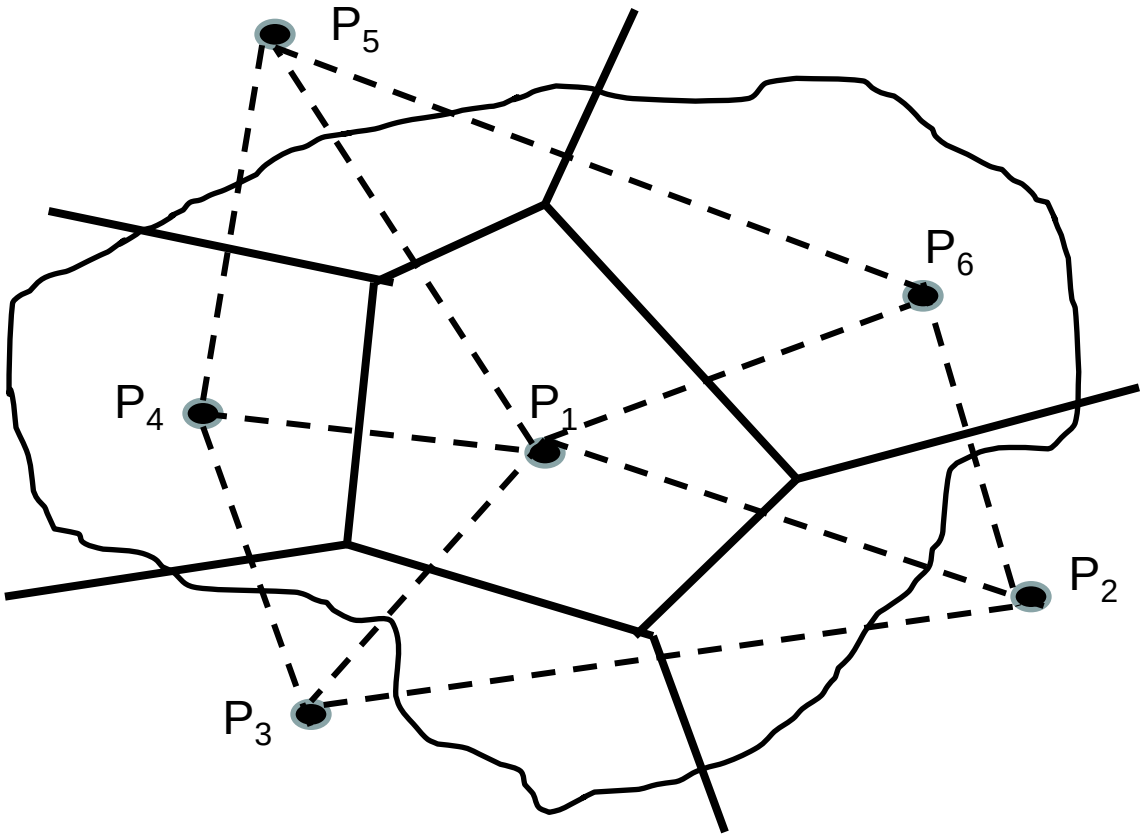
a. Drawing lines to connect each gauge

Thiessen polygon



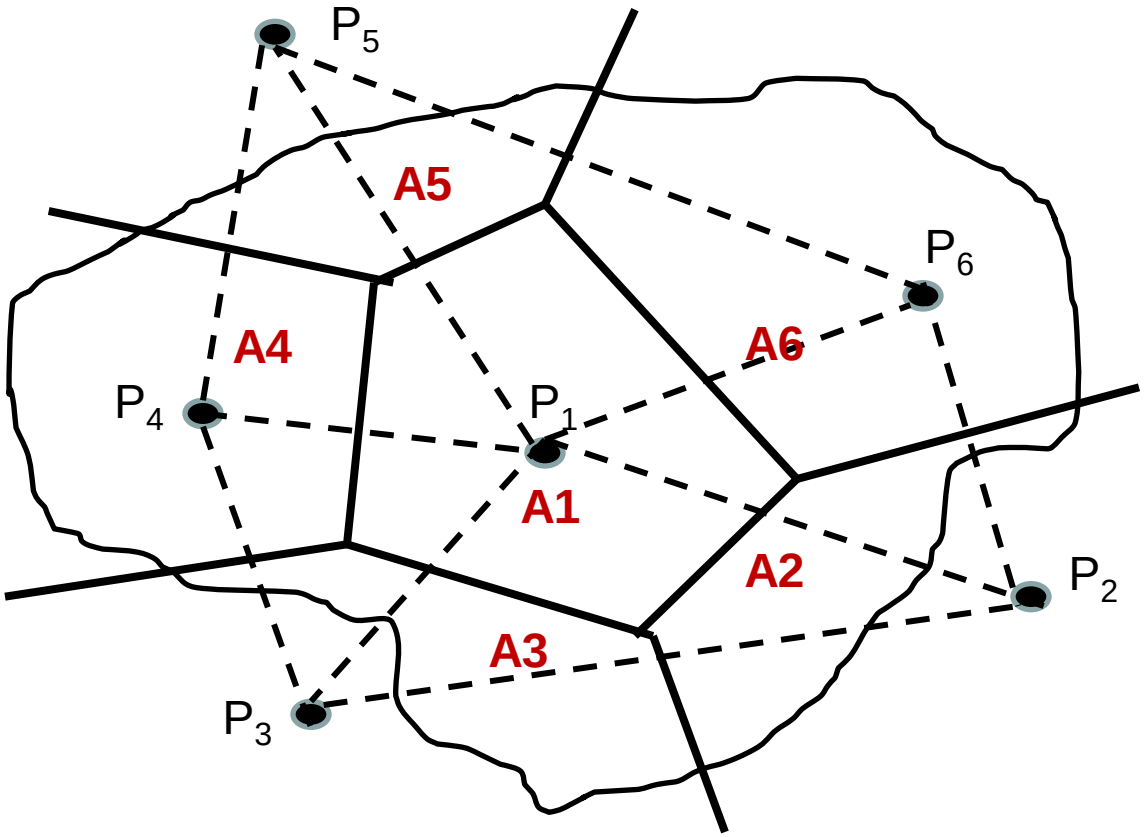
b. Then draw perpendicular bisector on each connecting line

Thiessen polygon



c. Connect bisectors to form polygons

Thiessen polygon



d. Determine area for each polygon

Arithmetic Average and Thiessen polygon

Rain gauge	**Precipitation (P_i)(mm)**	**Area of influence (A_i)(ha)**	**P_iA_i**
P_1	56	260	14560
P_2	40	58	2320
P_3	44	50	2200
P_4	52	250	13000
P_5	48	62	2976
P_6	52	290	15080
Total		970	50136

$$Pavg = \frac{\sum_{i=1}^{n} P_i}{n}$$

$$Av = \frac{292}{6} = 48.67$$

$$Pavg = \frac{\sum_{i=1}^{6} A_i P_i}{A_T}$$

$$Av = \frac{50136}{970} = 51.69\ mm$$

3. *Isohyetal Method*

- Construct isohyets between gauges
- Calculate area between isohyets
- Estimate average rainfall between isohyetals by applying same summation as Thiessen Polygon. Where A_i and P_i represent the area and average rain between two isohyetals.

Missing data points or rainfall at ungauged locations (1/5)

1. Simple average of surrounding gauges
    1. Guages must be of about equal distance from point of estimate
    2. Guages should have fairly equal rainfall at known gauges
2. Normal Ratio Method for missing data estimation
    1. When surrounding gauges differ by greater than 10% in measured rainfall
    2. Stations are weighted based on normal annual precipitation ratio’s

Missing data points or rainfall at ungauged locations (2/5)

Normal Ratio Method for missing data estimation

$$P_x = \frac{1}{k}\left(\frac{N_x}{N_1}P_1 + \frac{N_x}{N_2}P_2 + \ldots + \frac{N_x}{N_k}P_k\right)$$

Where:

N_x = normal long-term annual precipitation at location x (i.e. location of interest)

P_k = event precipitation at gauge k

N_k = normal long-term annual precipitation at gauge k

P_x = event precipitation at location x to be estimated

K = n-1

The event might be a single storm total, daily total, monthly total or annual value.

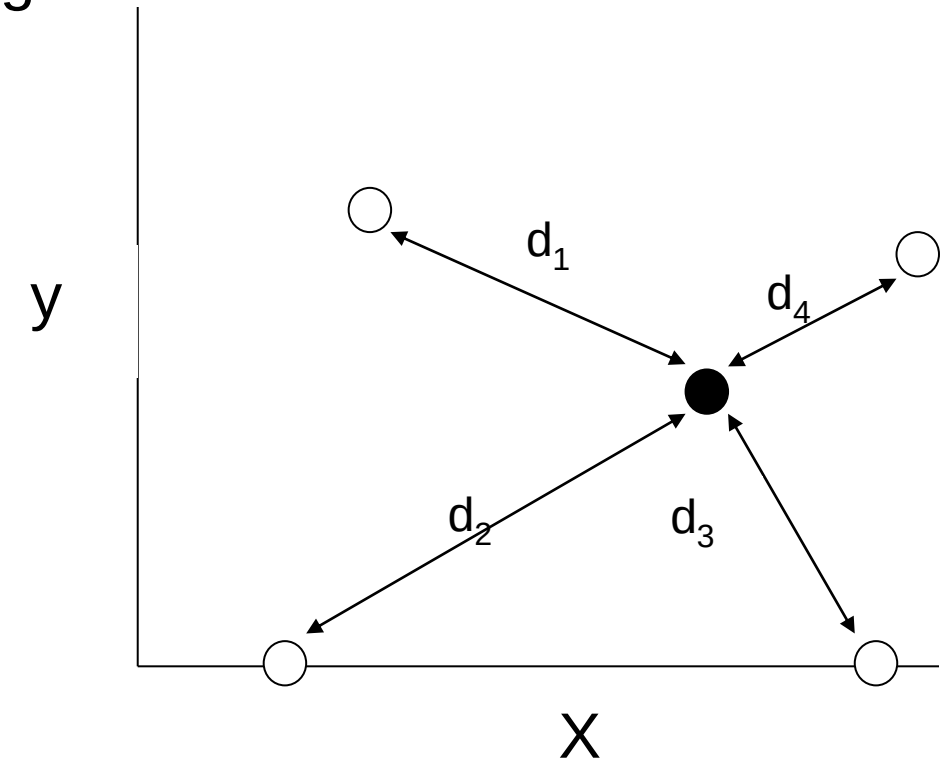
Example: Estimate the missing data for June 2010 at P_1

Rain gauge	Monthly rainfall for June 2010(mm)	Long term mean annual rainfall (mm)
P_1	x	1368
P_2	156	1324
P_3	149	1300
P_4	158	1348
P_5	148	1288
P_6	152	1290

Missing data points or rainfall at ungauged locations (3/5)

Reciprocal (Inverse weighted) Distance Method used for :

- missing data
- ungauged location data estimation



Missing data points or rainfall at ungauged locations (4/5)

Reciprocal Distance Method (2/3)

Principle: Surrounding gauges are weighted based on reciprocals of the sum of squares of distance away from the point of interest

1. Distance away from the point of interest

$$d_i = \sqrt{x^2 + y^2}$$

2. X and y are distances from the ungauged (missing data) location where precipitation is to be estimated

Missing data points or rainfall at ungauged locations (5/5)

Reciprocal Distance Method (3/3)

$$W_i = \frac{1}{d_i^2}$$

$$P_x = \frac{\sum_{i=1}^{n} P_i W_i}{\sum_{i=1}^{n} W_i}$$

Effective rainfall

- This is defined differently for the agriculturist, surface and groundwater hydrologist.
- For the agriculturist it is defined as that portion of total rainfall which is directly or indirectly useful for crop production. It must take care of the consumptive use by crops, land soaking or wetting for land preparation, leaching of salts.
- The effectiveness of rainfall depends on:
  1. Rainfall amount and intensity
  2. Evaporative demand
  3. Land and soil characteristics
  4. Antecedent soil-water status
  5. Groundwater status
  6. Farm and crop management practices
  7. Crop characteristics

Effective rainfall

- Effective rainfall is difficult to estimate because:
  - changes in infiltration rate with time
  - spatial variation of soil conditions
  - spatial and temporal variability of rain
- Forecasting is even more difficult because of the additional challenges due to:
  - uncertainty associated with rainfall pattern
  - variation in rainfall amount.

Effective rainfall measurement or estimation

Effective rainfall can be measured directly or estimated using empirical equations.

Direct measurement:

- Soil moisture change
- Daily soil moisture balance
- Integrating gauge
- Ramdas and
- **Lysimeter**

The empirical equations include:

- Renfro equation
- U.S Bureau of Reclamation method
- USDA-SCS method and
- **Potential ET/ precipitation ratio**

Effective rainfall measurement or estimation

- The Lysimeter method of measurement and potential ET/precipitation method of estimating effective rainfall are considered more accurate than the others.
- The effective rainfall values used for irrigation system design and water scheduling varies in different tropical countries.